Recent Interview Questions

1. What is your role and responsibility in your current or previous project?

Answer:
I worked as a DevOps Engineer. My tasks included managing cloud infrastructure, setting up CI/CD pipelines, using Docker and Kubernetes, writing automation scripts with Terraform, and monitoring systems. I also supported application deployments.

2. What AWS services have you used in your project and how?

Answer:
I used EC2 for virtual machines, S3 for storing files, RDS for databases, VPC for networking, IAM for user access control, CloudWatch for monitoring, and Lambda for serverless tasks. I managed them mostly using Terraform.

3. How would you rate your Linux skills?

Answer:
I would rate myself 7 or 8 out of 10. I can work well with the terminal, run system commands, write shell scripts, and handle file permissions, services, and logs.

4. What are some common Linux commands you use regularly?

Answer:
I use commands like ls, cd, top, ps, grep, chmod, chown, df, and journalctl. These help me manage files, check performance, find logs, and handle user permissions.

5. How can you check CPU usage on a Linux system?

Answer:
I use top, htop, or mpstat to check CPU usage. top shows live usage of CPU, memory, and processes. htop is more user-friendly.

6. How would you find the last 5 users who logged into a Linux machine?

Answer:
I can use the last command to see recent logins. To see the last 5 users, I run:
last -n 5

7. How would you containerize an Nginx web application using Docker?

Answer:
I would create a Dockerfile with nginx as the base image, copy the website files into the container, and expose port 80. Then I'd build and run the Docker container.

8. You have a Node.js app on port 8081 — how would you write a Dockerfile for it?

Answer:

```
FROM node:18
WORKDIR /app
COPY . .
RUN npm install
EXPOSE 8081
CMD ["node", "index.js"]
```

9. What is a multi-stage Dockerfile and why is it useful?

Answer:
A multi-stage Dockerfile helps reduce image size. The first stage is used to build the app, and the second stage copies only the final output. Yes, we can use the second stage as the base if needed.

10. Write a Dockerfile for a ReactJS application.

Answer:

```
FROM node:18 as build
WORKDIR /app
COPY . .
RUN npm install
RUN npm run build

FROM nginx:alpine
COPY --from=build /app/build /usr/share/nginx/html
```

```
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

11. Have you worked with Kubernetes? Can you describe your experience?

Answer:
Yes, I have deployed applications using Kubernetes. I created Deployments, Services, ConfigMaps, and used kubectl to manage clusters. I also worked with Helm charts and monitored pods.

12. Based on a Dockerfile, how would you write a Kubernetes Deployment manifest?

Answer:
I would create a YAML file with apiVersion, kind: Deployment, and define the spec with replicas, container image, ports, and labels. Then apply it using kubectl apply -f.

13. What are the types of services in Kubernetes and when would you use each?

Answer:
There are 4 main types:

- ClusterIP (default, internal access)
- NodePort (exposes service on a static port on each node)
- LoadBalancer (uses cloud load balancer)
- ExternalName (maps service to an external DNS)

14. What is a headless service in Kubernetes and what's its use case?

Answer:
A headless service doesn't have a cluster IP. It's used when we want DNS resolution without a load balancer, such as in StatefulSets or when each pod should be accessed directly.

15. A pod is stuck in a CrashLoopBackOff state — how would you troubleshoot this?

Answer:
I would check logs using kubectl logs <pod-name>, then inspect the pod

using kubectl describe pod. It usually happens due to app errors or failed dependencies.

16. In a Kubernetes cluster, one ReplicaSet is not functioning properly. How would you debug it?

Answer:
I would describe the ReplicaSet and its pods using kubectl describe rs <name>. Then check if the pods are scheduled, healthy, and if there's any issue in the events section.

17. How does a Kubernetes application communicate with the external world?

Answer:
We expose the app using a Service of type LoadBalancer or NodePort. If it's behind an Ingress controller, we use an Ingress resource for routing external traffic.

18. What are the lifecycle states of an EC2 instance?

Answer:
The main states are:

- Pending
- Running
- Stopping
- Stopped
- Shutting-down
- Terminated

19. What is VPC Peering, and how does it work across regions or accounts?

Answer:
VPC Peering connects two VPCs so they can communicate privately. It works across regions and accounts using peering connections, route tables, and proper security group rules.

20. How do you connect to an EC2 instance inside a private subnet?

Answer:
I use a bastion host in a public subnet. First, I connect to the bastion host using SSH, then connect to the private EC2 instance from there.

21. What's the difference between Security Groups and NACLs in AWS?

Answer:
Security Groups work at the instance level and are stateful. NACLs work at the subnet level and are stateless. NACLs have separate rules for inbound and outbound traffic.

22. How would you set up VPC Peering between two VPCs?

Answer:
Create a VPC peering connection, accept it in the other VPC, update route tables in both VPCs, and allow traffic in security groups and NACLs.

23. What is the AWS Shared Responsibility Model? Can you explain it with an example?

Answer:
AWS handles security *of* the cloud (hardware, network), while we manage security *in* the cloud (data, access, configurations). For example, AWS secures the servers, but we must set up IAM roles properly.

24. How does S3 manage versioning and object locking for data durability?

Answer:
Versioning keeps all versions of an object. Object locking prevents deletion for a set time. These features help protect data from accidental loss or overwrite.

25. How would you monitor the performance of an EC2 instance using CloudWatch?

Answer:
I check metrics like CPU, disk, memory, and network usage. I also set alarms to notify when usage is too high or something goes wrong.

26. How can you use CloudWatch to track application metrics on EC2?

Answer:
Install and configure the CloudWatch agent. It collects logs and custom metrics like memory or app-level data. Then I view them in the CloudWatch dashboard.

27. Can you write a Terraform script to provision an EC2 instance and an S3 bucket?

Answer:
Yes. I define resources like aws_instance and aws_s3_bucket in Terraform, then use terraform apply to create them. I also use variables for reusability.

28. Write Terraform code to create 5 EC2 instances, each with a unique name and instance type.

Answer:

```
variable "instances" {
 default = {
  "web1" = "t2.micro"
  "web2" = "t2.small"
  "app1" = "t3.micro"
  "app2" = "t3.small"
  "db1"  = "t3.medium"
 }
}

resource "aws_instance" "multi" {
 for_each = var.instances
 ami = "ami-12345678"
 instance_type = each.value
 tags = {
  Name = each.key
 }
}
```

29. What CI/CD steps do you follow during software development?

Answer:
I start with code push to Git. A pipeline runs tests, builds the app, creates a Docker image, and pushes it to a registry. Then it deploys to a test or production environment.

30. How do you build and push a Node.js Docker image to Docker Hub using a CI/CD pipeline?

Answer:
In the pipeline, I add steps to build the Docker image using the Dockerfile, tag it, log in to Docker Hub, and push it using docker push.

31. How would you connect AWS Lambda with API Gateway for serverless deployment?

Answer:
I create a Lambda function and an API Gateway endpoint. Then I connect the API to the Lambda function so that HTTP requests trigger the function.

32. Do you have experience working with RESTful or GraphQL APIs? How have you used them?

Answer:
Yes. I've used REST APIs to integrate services, test them using Postman, and monitor responses. I also deployed backend services that exposed REST endpoints for frontend teams.

1. Introduction
2. What is AWS and what type of services you have used in AWS.
3. What is a role and user in AWS? Also, tell difference between them.
4. How can you create a s3 bucket from CLI and also how can you view the list of s3 buckets in AWS.
5. How you will add a role to a specific service in the AWS let's take EC2.
6. Your jenkins service is not running what troubleshooting steps would you follow.
7. Name some of the plugins you have used in your projects.
8. Explain about the ci/cd stages?
9. Name some of the default variables in jenkins.
10. What is the port number for nexus? How will you store the war files in the nexus repo.
11. What type of repository you are using, either group repository or a standard repository.

12. What is the command to print env variables in Linux, and also what are some default env variables.
13. What is file operator.
14. If a user is created what must be his default path?
15. What is the command to create a user and group in Linux.
16. What type of Linux distris you have worked with and what is the command to see the current Linux distro.
17. If you enter a command Hostnamectl what will be the output.
18. In Linux user is created there will be few default files created what are they, and also where the user details will be stored in Linux.
19. Linux command to see the disk free space.
20. Write shell script to print largest number out of 3 numbers.
21. What is the command to run the bash script.
22. What is the default Linux directory and file permissions.
23. How you will pass arguments in Linux. Write a bash script to pass the arguments.
24. What is #!/bin/bash what happens if we don't use this in the shell script.
25. What is diff between soft link and hard link and commands to create them
26. What is the command to run a python.py file in Linux.
27. What is docker?
28. What is the difference between CMD and ENTRYPOINT
29. Suppose you are running some process in container how can you see from Linux server without going into the container.
30. Command to create docker container using an existing image for example nginx.

2nd round Technical interview questions(f2f)

Brief Introduction
1. A user don't have the execute permissions for an script file. What are the ways you will check and assign.
2. A jenkins service is not running in the instance what steps would you follow?
3. How would you troubleshoot the Disk space issue in Linux explain in details with steps?
4. What type of file permissions are there what are they?
5. can't able to SSH into a server what troubleshooting steps would you follow?
6. Linux server is not coming up after reboot what steps would you follow?
7. Jenkins service is not able to access from chrome, what would be the reason?
8. Where the docker logs stored in the Linux?
9. What is a terrafom state file and backend?
10. Write terraform code for an instance launch, also use variables.
11. How would you manage terraform state file in a team environment?
12. Terraform file is corrupted how would you troubleshoot?
13. Docker architecture explain?
14. Jenkins plugins, what is the purpose of pollSCM & Build periodically?
15. How would you build a jenkins pipeline manual and automatic explain?
16. Jenkins pipeline explain, and also write a basic pipeline syntax?

17. How we can store the credentials in jenkins securely and call them in the pipeline?
18. What type of RDS you have worked with?
19. Basic sql commands on MYAQL?
20. How would you push the images to ECR?
21. What is meant by prometheus & grafana?
22. In which linux file we will mention the prometheus configs to get alerts?
23. Where the logs will be stored for grafana and prometheus in linux?
24. what is composefile and swarm?
25. A crontab is failing how would you troubleshoot?

Client round interview questions

1. Introduction, project explanation, roles & responsibilities.
2. What is an EC2 and how do you launch it from UI and through code, explain?
3. If a server is down what are your troubleshooting steps?
4. How do you rate yourself on linux?
5. Describe the commands that are used for checking RAM, ROM, CPU usage, Memory. What is the command for checking the running processes on Linux.
6. Explain about the importance of filesystem? What type of filesystem you are using?
7. What is the path for storing the logs in linux? What type of logs will be store in the /var/log/messages?
8. Command to create user & Tell me some files which are in /etc directory.
9. Expalain about the entries /etc/passwd, /etc/shadow, /etc/groups files?
10. Where does the jenkins logs stored in the linux?
11. Suppose jenkins service is not running, what troubleshooting steps will you follow.
12. what is terraform? What is a terraform statefile, how do you manage terraform statefile it in the team environment.
13. Terraform statefile corrupted how do you troubleshoot?
14. Explain about terraform provider block & resource block, terraform lifecycle.
15. Which type of S3 storage areyou using currently & to reduse the cost which type of s3 will you use.
16. Suppose the terraform is running in the version 5.12 and whenever we apply changes in the code and run it, will the terrafrom runs on 5.12 or will it take the latest version.
17. write terraform simple code for s3 backend.
18. What is ansible? Explain the architecture of it and it's use case in the production.
19. Where we can store the secrets in the ansible?
20. Explain how you can secure the yml file and secrets files in the ansible?
21. Can we changes the ansible-vault password? If yes what was the command to change?
22. Tell me some of the ansible playbooks.yml file that you have used in the production.
23. Explain about the /hosts & /roles in ansible?
24. What is jenkins and why we are using it?
25. Tell me the jenkins stages that you have worked with?
26. what is the plugin that we used to run muliple branches in the git?
27. Name some plugins you have worked with
28. Jenkins is pipeline running slow and how do you troubleshoot it?

29. How can you change the default password in jenkins.
30. Where do you store the passwords and usernames of sonarqube or nexus in jenkins securly?
31. How will you run the stages parellel in jenkins?
32. Default port number of jenkins?
33. What type of parameters are there in jenkins?
34. How do you troubleshoot if the jenkins.service is down? And tell the commands to start, restart or enable jenkins service?
35. What is git and why we are using it?
36. What is git rebase & git merge?
37. What are your branching stratagies in git?
38. What is git stash?
39. Tell me some commands you know in git?
40. Do you have any idea on github actions, id yes what it means & how it works?
41. What type of services you have worked with in AWS?
42. To establish a connection between the service to service, how do you acheive it?
43. What type of RDS you have used in your current project
44. What is lambda and why we are using it?
45. What is DNS & route53?
46. What are your achievements in your current organisation?
47. Have you learnt any tool or anything on your own recently?
48. client Explained about his project, roles and then work timings.

Festo India – Round 1 DevOps Interview Questions

What is CI/CD?

Write a Python script to create an EC2 instance and enable monitoring.

How do you integrate SonarQube into a Jenkins pipeline? (Explain step-by-step)

Suppose no team members are available and you need to create the whole infrastructure from scratch — from setup to deployment and monitoring — how would you achieve this?

Do you have any knowledge or experience in migration from one cloud or infrastructure to another? Explain.

How do you troubleshoot a Jenkins pipeline if it fails?

What branching strategy do you follow in your project?

Write a sample YAML file in Kubernetes (any resource of your choice).

Do you have knowledge or experience managing VMs and virtual machines? Explain.

Explain your project CI/CD pipeline in detail.

What is a Jenkins runner? Explain its role.

In our project/organization we use GitLab CI — since you have experience with Jenkins, how comfortable are you adapting to new CI/CD tools or technologies?

1.Explain You Project Infrastructure
2. When user hit the your URL What will happen..!?
3. How Route 53 will connect..!?
4. difference between the Git fetch/ pull...!?
5. What is Git Stach...!?
6. What is Git Cherrypic...!?
7. How to reduce the Docker file size..!?
8. What is the difference between the COPY / ADD...!?
9. What is the difference between the CMD / Entrypoint...!?
10. Did you initialized the SonarQube in your CI/CD pipelines...!?
11. How did you initialized/ configured...!?
12. What is the Backend...!?
13. If one instance is deleted unexpectedly how we will recover with Terraform...!?
14. what is provisions...!?
15. What is providers...!?
16. Can we configure the on prem server in Terraform...!?
17. If we configure how will you..!?
18. What is your Roles and responsibilities..!?
19. What is the Kubernetes components...!?
20. Explain the kubernetes structure..!?

Deployment strategy

What is the procedure for Deployments

How do u deploy the same app in sit uat prod

Secure the data

Did u worked on automation scripts for desktop apps or any

While configuring tools what is the issue u faced

Tell me about ur project

What is the method u follow for prod

Branching strategies , recommend a best Branching strategy

Gradle experience

What is the major issue u faced while deployments

What kind of applications u usually worked on like exactly what

Do u directly deploy to prod after sit uat completed

LTI

Deployment strategy

What is the procedure for Deployments

How do u deploy the same app in sit uat prod

Secure the data

Automation scripts for desktop apps

While configuring what is the issue u faced

Tell me about ur project

What is the method u follow for prod

Branching strategies

Gradle experience

What is the major issue u faced while deployments

What kind of applications u usually worked on like exactly what

Do u directly deploy to prod after sit uat completed

Wipro ===

Entire pipeline set up end to end pipeline

Aws vpc , iam Roles, ec2 ecs

Java end to end Integrate end to end pipeline using aws terraform

Harness deployments code

Imperative vs declartive

How to integrate tools pipeline in jenkins

Explanation pom.xml along with what are the files exists in pom

Docker containerization

Git vs github

Github actions

Git merge Git rebase

Docker File Components

Docker namespaces

Mphasis ...

Self introduction

Declaritive pipeline vs scripted pipeline

Mvn clean install mvn deploy mvn package

Parallel pipeline

Blue green deployment

Health checks

Master slave architecture

Shell scripting *

Lloyds

List running process

How to edit file and save in linux

Blue green deployment

Softlink and hardlink

Impetus Technologies:

Round-1:

1. How to automate a process using pipeline?
2. Types of errors you will face in pipeline?
3. How to create resources in terraform?
4. How to delete unused containers in docker?
5. Write a command to dick utilisation of a server?
6. Write a script to find disk utilisation of a container?
7. How to integrate pod monitoring tools in Jenkins pipeline?
8. How to get notifications about pipeline?
9. Tell me the difference between deployment and stateful set?
10. What is use case for deployment and stateful set?
11. What type of deployment strategy you used in your project?
12. How to deploy an application in production?

Round-2:

1. How do you deploy an application in production environment? Explain.
2. What type of issues you faced while deploying an application in production environment? How did you overcome those issues?
3. When ever you updated the application with new deployment. how did you handle/route the traffic to new version application?
4. How you maintain code in Github?
5. Explain deployment strategy you used in your project?
6. How do you automation deployment?
7. How do you implement argoCD in deployment?
8. What is major issue in production you have faced in last 3 to 6 months?

EY:

Round-1:

1. How do you automate process using Jenkins pipeline?
2. How do you use (integrate) tools in Jenkins pipeline?
3. How do pass variables to pipeline?
4. How do you pass credentials in pipeline?
5. What is SCM?
6. What is use of branches?
7. What is merge and rebase? What is difference between merge and rebase?
8. How do you bring git hub code to local?
9. What is git push?
10. What is git stash?
11. Who do you trigger pipeline when code is committed in git?
12. How to create a file in Linux?
13. Write a command to create a directory?
14. How to change file permissions?
15. How to create user?
16. How to change owner to a file?
17. Give me the value for read permission and write permission?
18. What is ansible?
19. What is use of playbooks?
20. Write a playbook to install 3 services.
21. What is Kubernetes?
22. Explain Kubernetes architecture.
23. What is purpose of scheduler?
24. What is deployment? Explain a use case.
25. What is stateful set? Explain use case.

26. What is difference between deployment and stateful set?
27. What is docker file?
28. Explain docker components.
29. Explain docker file components.
30. What is difference between COPY and ADD?
31. What is nginx port number?
32. Write a docker file to a web-server and the image should be light weight.
33. What is docker volume and how do you attach a volume to a container?
34. Explain terraform. Tell me the command in terraform
35. What is state file?

1) Self-Introduction
2) In your project, What Linux commands will you use when you start your day ?
3) How do you check live system logs or errors?
4) How do you check network connectivity between servers?
5) How to integrate jenkins with ansible?
6) If jenkins build fails, how will you trobuleshoot or resolve?
7) What jenkins stages you implemented in your project?
8) What are DDL & DML commands in SQL?
9) How will you find duplicate records in a table?
10) What are the roles and gathering facts in ansible?
11) What is idempotency
in ansible?
12) What deployment strategy you implemented in your projects?
13) What types of automation scripts you wrote and explain them
14) What challenges you faced while deployment and how you resolved?
15)Are you familiar with dynatrace and autosys?
16) What monitoring tools you worked and how did you integrate them with kubernetes?

How do you establish and check the connection between two servers.
Your server is not starting properly, how do you debug that.
What are the playbooks you have written.
Ansible vault
What type of Jenkins pipeline you built
What are the stages in it.
What is meant by a job in Jenkins
Your jenkins job is failing, how do you debug.
Why kubernetes?
How do you debug a Pod which is going to error state
How do you check/fetch error logs in a server
In which location a server normally stores the logs
Which type of deployment strategy are you following and why
Write a shell script that you have written in your project

Git branching strategies
How do you integrate Prometheus with Grafana
Why splunk is needed.

Devops Interview questions
1.Explain Devops tools
2.Differences between Docker and Kubernetes
3.What are docker file components and write Docker file?
4.What is Docker swarm?
5.Whats is Docker Compose?
6.What is kubernetes?
7.Differences between statefulset and Deployment?
8.what you mean by pod notready state?
9.What is ingress
10.Tell me about confimaps and Secrets?
11.command for disk space
12.Command to check memory space
13.If your pod is starting continuously what you will do?
14.Command to go inside the pod?
15.What is 777 permission means?
16.What is the command to give write permission
17.what will be the command to check the largest file in the directory?
18.What is TLS?